

Yukti

Numbers of record + thresholds

No GPU

Stage 2 PoC - 1-2 Sept

SIH26104 - Team Sonix

Your mission: Own every number we report, and give us the alarm levels that actually work on real audio. No GPU needed.

THE SITUATION - READ THIS FIRST

Where we are. SONIX listens to a call in 4-second chunks and shows green / amber / red. On the academic benchmark it is strong: **1.49% EER** on ASVspoof-2019 LA. But on our own real recordings it wrongly flags roughly **half of genuine voices as fake**.

Why. The model only ever learned from clean studio recordings of real people - same good microphone, same quiet room. Our phone and laptop recordings sound nothing like that. So the model thinks: "this is not like any real voice I know, therefore fake." **The model is not broken. It is narrow.** It has never heard normal life.

The fix - three things. (1) **Show it more kinds of real:** add genuine recordings from many phones, mics, rooms and languages to training. (2) **Make it ignore silence:** quiet gaps are what break the "minimal voice" clips. (3) **Reset the alarm levels:** our amber/red numbers came from the clean test, so we re-set them using real recordings.

HARD RULE: we never train on In-the-Wild or on any eval set. Those are our honest test. Training on them would destroy the one number that proves we generalise.

THE PLAN - THREE LANES RUNNING AT ONCE

LANE 1 - FIX THE MODEL

Silence gate, more real audio, RawBoost augmentation, retrain, recalibrate. This is what stops the false alarms.

Yugal - Akshat - Yukti

LANE 2 - HONEST NUMBERS

2021-DF EER, In-the-Wild EER, codec 2x2, metrics + calibration. These are the numbers we quote to the jury.

Navya - Anubhav - Yukti

LANE 3 - A DEMO THAT CANNOT FAIL

Replay mode on validated clips, clip library, pitch, Q&A, backup video. The demo never depends on live luck.

Suryansh - Anubhav - Akshat

YOUR TASKS

P1 TONIGHT

Finalise metrics.py

- One place that computes EER, so our numbers never disagree.
- Positive class is spoof = 1. Match the cross-check already inside train.py.
- Also produce a DET curve and a score-distribution histogram as PNGs for the slides.

```
eer(labels, scores) # 1 = spoof
det_curve(...) -> det.png ; hist(...) -> scores.png
```

Done when: Anyone can reproduce every reported number with metrics.py.

P2 TONIGHT

Temperature-scaling calibration

- A direct PPT promise. Fit one temperature value on dev logits so the 0-1 output is a real probability, then apply it to eval scores.
- Calibrated scores are what make thresholds meaningful across datasets.

```
calibration.py: fit T on dev logits -> apply to eval
```

Done when: Calibrated scores produced and saved.

P1
TOMORROW

The real-world operating point - our biggest fix

- As soon as Yugal scores Akshat's labelled real/fake clips, find the amber and red thresholds that ACTUALLY separate them.
- Our current 0.10 / 0.90 came from clean studio data and is far too trigger-happy on real audio.
- Give Yugal two numbers and a plot showing the separation.

Done when: Two threshold numbers handed to Yugal, with a plot.

P2
TOMORROW

Results table for the slides

- One table, every number: 2019 LA (baseline/aug/robust), codec 2x2, DF21, In-the-Wild, plus the real-clip false-positive rate before and after our fix.
- The before/after is our strongest slide - it shows we found a problem and fixed it.

Done when: A single results table ready to paste into the deck.

YOUR CHECKLIST

- ☐ metrics.py is the single source of truth
- ☐ calibration.py done
- ☐ Real-world amber/red thresholds delivered
- ☐ Results table complete

Note: You are blocked only on scored CSVs from Yugal. Everything else you can start now.

WORDS WE KEEP USING

EER	One number for how wrong the model is. Lower = better. 1.49% is strong. 50% = coin flip.
bonafide	A real human voice.
spoof	An AI-cloned / synthetic voice.
embedding	The 1024-number fingerprint of a 4-second audio chunk.
head	The small model we actually train. The big wav2vec2 front-end is frozen (never trained).
VAD	Voice Activity Detection - is anyone actually speaking in this chunk?
augmentation	Making altered copies of training audio (phone-compressed, noisy) so the model sees variety.
calibration	Making the 0-1 score mean a real probability, so thresholds are meaningful.
domain shift	Test audio looks different from training audio. THIS is our core problem.